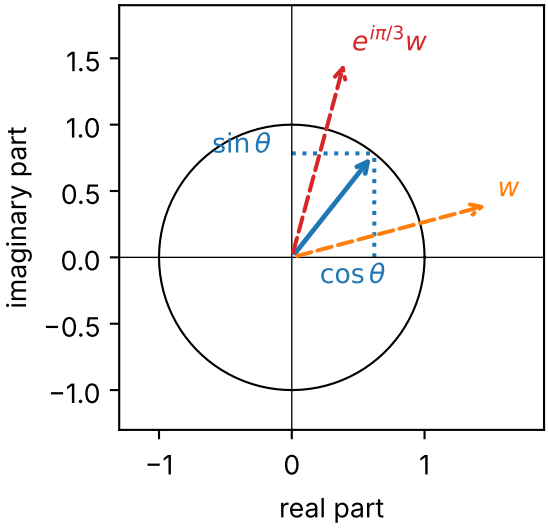
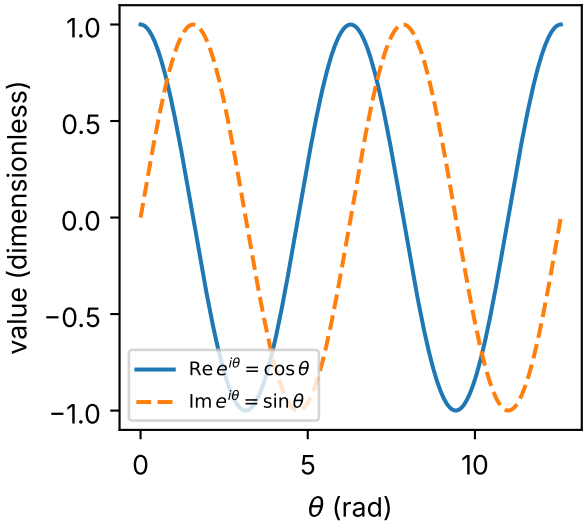

$e^{i\pi/3}w$
w
$\sin\theta$
$\cos\theta$
imaginary part
real part
1.5
1.0
0.5
0.0
−0.5
−1.0
−1
0
1


value (dimensionless)
θ (rad)
$\mathrm{Re}\, e^{i\theta} = \cos\theta$
$\mathrm{Im}\, e^{i\theta} = \sin\theta$
1.0
0.5
0.0
−0.5
−1.0
0
5
10